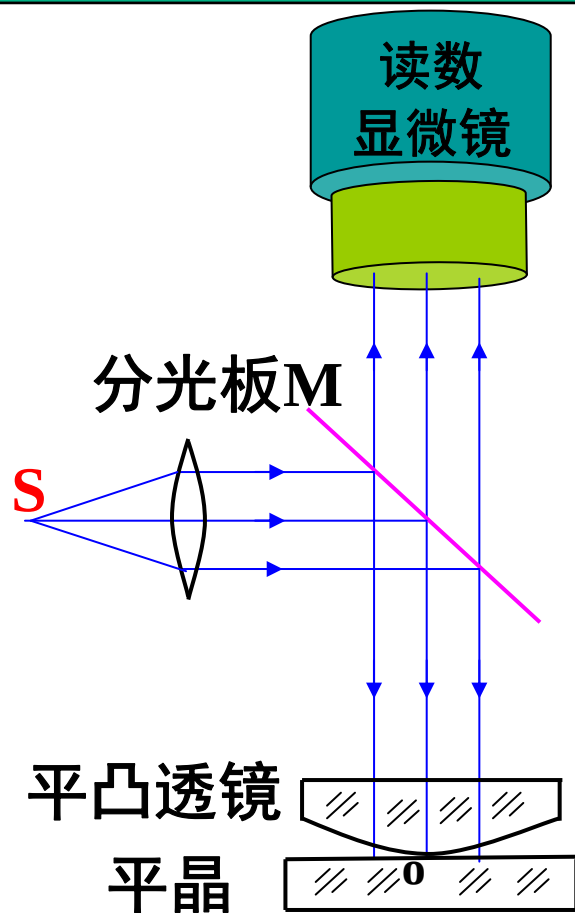
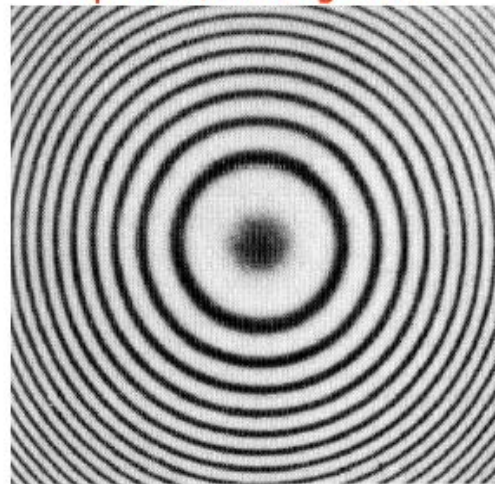


§ 3.3.3 牛顿环

实验装置



top view, looking down:



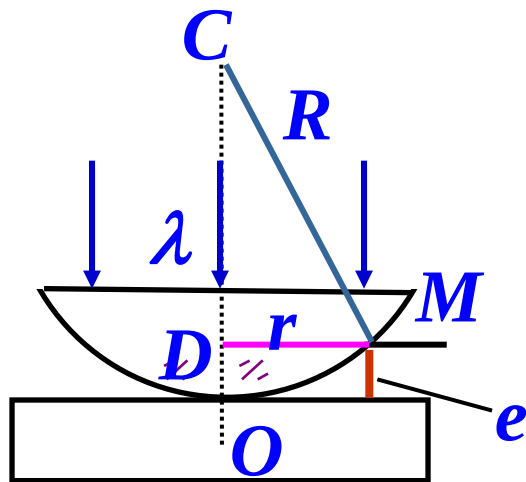
干涉条纹分析

$$\delta = 2e + \frac{\lambda}{2}$$

$$= k\lambda, \quad k = 1, 2, 3, \dots \quad \text{明纹}$$

$$= (2k' + 1)\frac{\lambda}{2}, \quad k' = 0, 1, 2, \dots \quad \text{暗纹}$$

O 处, $e = 0$, 暗斑

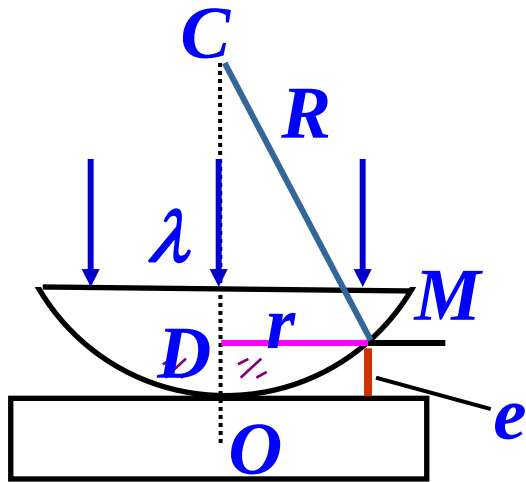


明暗环半径（反射光中）

$$r^2 = R^2 - (R - e)^2 \approx 2R e$$

暗环: $\delta = 2e + \frac{\lambda}{2} = (2k + 1)\frac{\lambda}{2}$

$$e = k \frac{\lambda}{2} \quad r^2 = 2Rk \frac{\lambda}{2}$$



$\Rightarrow$ 第 k 个暗环半径 $r_k = \sqrt{kR\lambda} \propto \sqrt{k}$

明环

$$\delta = 2e + \frac{\lambda}{2} = k\lambda \quad (k = 1, 2, \dots)$$

$$e = \left(k - \frac{1}{2}\right) \frac{\lambda}{2} \quad r^2 = 2Re = 2R\left(k - \frac{1}{2}\right) \frac{\lambda}{2}$$

⇒第 k 个明环半径

$$r_k = \sqrt{\frac{(2k-1)R\lambda}{2}}$$

讨论:

1) 花纹为以触点O为圆心的明暗相间的圆，从中心向外干涉级次越来越高

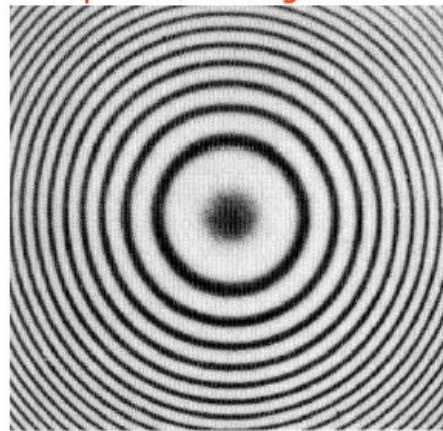
2) 条纹内疏外密

相邻暗环半径差 $\Delta r = r_{k+1} - r_k$

$$= \sqrt{(k+1)R\lambda} - \sqrt{kR\lambda}$$

$$= \frac{1}{\sqrt{k+1} + \sqrt{k}} \sqrt{R\lambda} \quad k \uparrow, \Delta r \downarrow$$

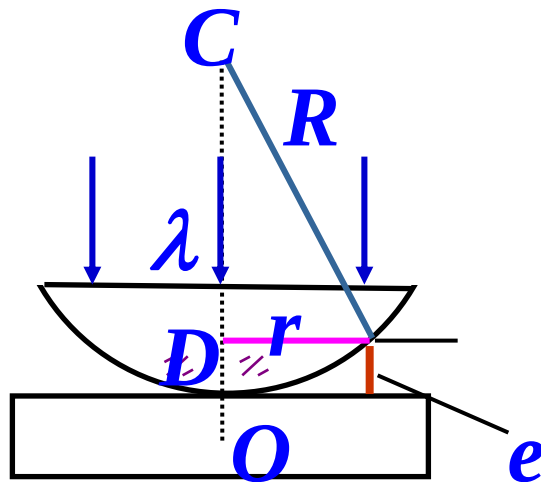
top view, looking down:



3)白光入射, 同一级条纹, 红色在外圈, 紫色在内圈

4)条纹动态分析

$$\delta = 2e + \frac{\lambda}{2}$$



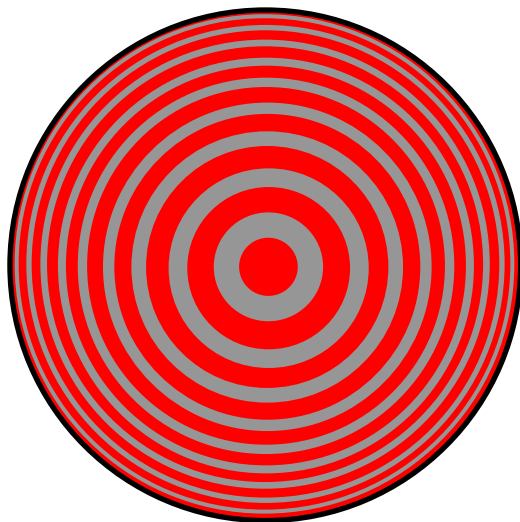
玻璃板下移，膜厚 $e \uparrow$ ，等厚线向中心收缩，
花纹也向中心收缩

厚度每改变 $\lambda / 2$ ，条纹向中心缩进一条。

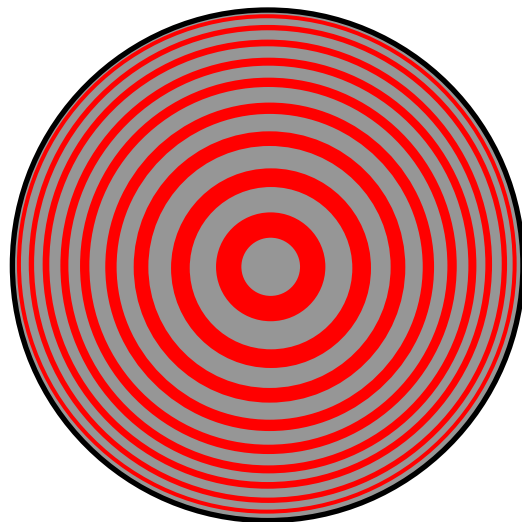
膜厚 $e \downarrow$ ，等厚线外移，

厚度每减少 $\lambda / 2$ ，条纹向中心冒出一条。

5) 透射光中观察牛顿环
无额外程差, O 点是亮斑



透射光牛顿环



反射光牛顿环

应用

$$r_k = \sqrt{kR\lambda} \quad r_{k+m} = \sqrt{(k+m)R\lambda}$$

实用的观测公式: $r_{k+m}^2 - r_k^2 = mR\lambda$

▲测透镜球面半径 R :

已知 λ , 测出 m, r_{k+m}, r_k , 可得 R

▲测 λ : 已知 R , 测出 m, r_{k+m}, r_k , 可得 λ

[例]: 用 $\lambda_1 = 6000\text{\AA}$ 和 $\lambda_2 = 4500\text{\AA}$ 的两种波长的光观察牛顿环, 用 λ_1 时的第 j 级暗圈与用 λ_2 时第 $(j + 1)$ 级暗圈重合, 透镜的曲率半径 $R = 90\text{cm}$, 求: 用 λ_1 时的第 j 级暗圈的直径。

解: 暗圈的半径

$$r = \sqrt{kR\lambda} \quad \sqrt{jR\lambda_1} = \sqrt{(j+1)R\lambda_2}$$

$$j = \frac{\lambda_2}{\lambda_1 - \lambda_2} = \frac{4500}{6000 - 4500} = 3$$

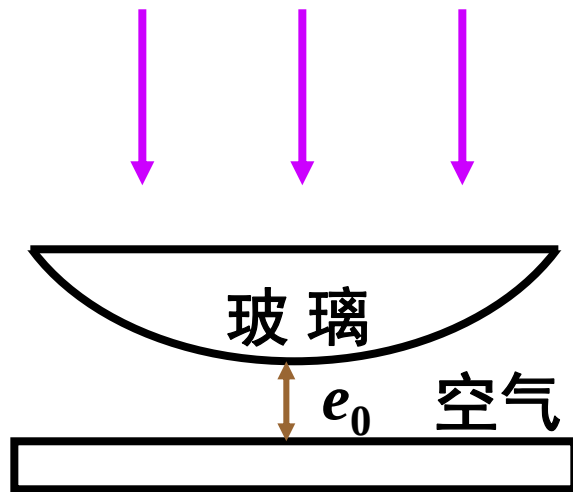
$$r_3 = \sqrt{3R\lambda_1} = \sqrt{3 \times 6000 \times 10^{-10} \times 90 \times 10^{-2}}$$

$$= 1.27 \times 10^{-3} \text{ (m)}$$

$$D_3 = 2 r_3 = 2.54 \times 10^{-3} \text{ (m)}$$

[例]：如图所示, 牛顿环装置的平凸透镜与平板玻璃有一小缝 e_0 . 现用波长为 λ 的单色光垂直照射, 已知平凸透镜的曲率半径为 R 。

求：反射光形成的牛顿环的各暗环半径



解：设某暗环半径为 r ,由图可知,根据几何关系,

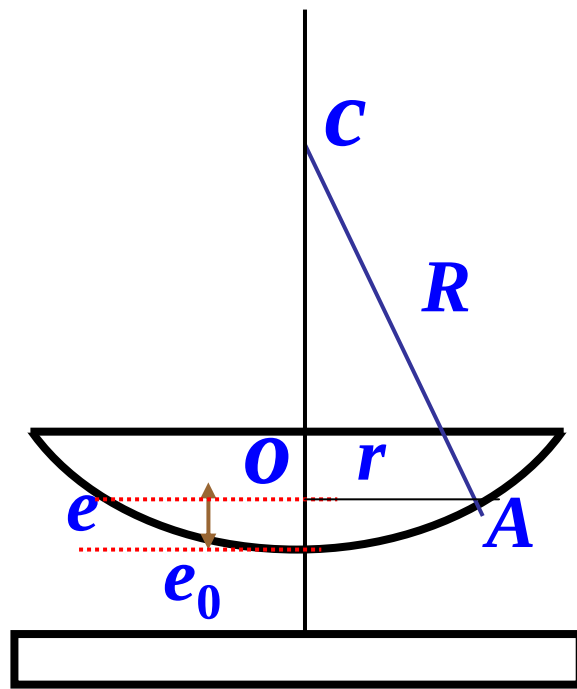
在直角三角形 Aoc 中,

$$r^2 = R^2 - (R - e)^2 = 2Re - e^2$$

因为 $R \gg e$,上式中可略去 e^2 ,

于是得

$$e = \frac{r^2}{2R} \quad \text{-----}(1)$$



$$e = \frac{r^2}{2R} \quad \text{-----(1)}$$

再由干涉减弱条件有：

$$2(e + e_0) + \frac{\lambda}{2} = \frac{1}{2}(2k + 1)\lambda \quad \text{-----(2)}$$

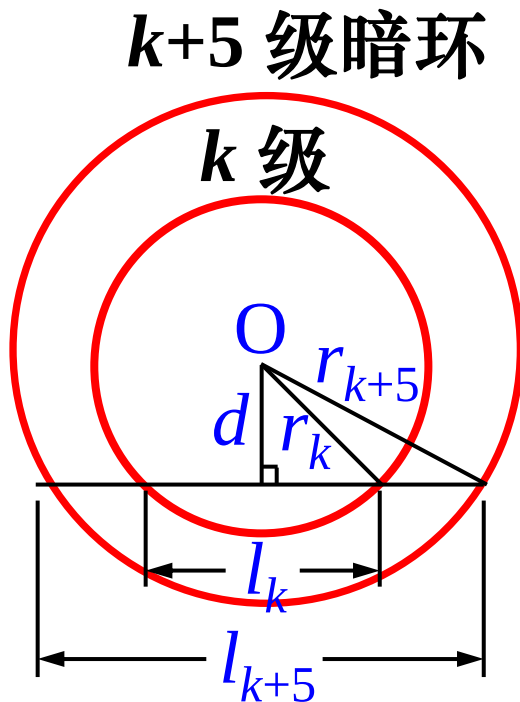
把(1)式代入(2)式可得

$$r = \sqrt{R(k\lambda - 2e_0)} \quad (k \text{ 为整数, 且 } k > \frac{2e_0}{\lambda})$$

**[例]:用波长为589.3nm的单色光观察反射光牛顿环
如图，第 k 和 $k + 5$ 级暗环弦长分别为 3.00mm，
4.60mm，求凸透镜曲率半径。**

解：暗环处满足光程差

$$\delta = 2e + \frac{\lambda}{2} = \frac{r^2}{R} + \frac{\lambda}{2} = (2k + 1) \frac{\lambda}{2}$$



$$\left. \begin{aligned} \therefore r_k^2 &= k\lambda R \\ r_{k+5}^2 &= (k+5)\lambda R \end{aligned} \right\} R = \frac{r_{k+5}^2 - r_k^2}{5\lambda}$$

$$\therefore r_k^2 = d^2 + \left(\frac{l_k}{2}\right)^2, \quad r_{k+5}^2 = d^2 + \left(\frac{l_{k+5}}{2}\right)^2$$

$$\therefore l_{k+5}^2 - l_k^2 = 4(r_{k+5}^2 - r_k^2) \quad \therefore R = \frac{l_{k+5}^2 - l_k^2}{20\lambda} = 1.03\text{m}$$